# Functional cognitive performance at age 53 years

Marcus Richards, Karen McKinnon, Imran Shah | April 2014

## Contents

## Document details

| | |
|---|---|
| **Categories of variables** | Cognitive Function |
| **Summary of work undertaken** | Conversion, cleaning and derivation of 1999 cognition data. |
| **Output variables** | recm99c, recy99c, intm99c, inty99c, wlcd99, wlt199, wlin199, vsrw99, vscl99, vsms99, vser99, wlt299, wlin299, nart99, wlt399, wlin399, wltx99, wlinx99, wlt499, wlin499, vsht99, anin, remem, wlt99, cansp99 |

## Summary

Cognitive function was first tested at age 8 years in a school setting, and again in this setting at ages 11 and 15 years. With the single exception of a test repeated at age 26 years (reading comprehension) cognitive function was not assessed again until age 43 years. At this time there was a shift away from tests of intellectual ability and towards the assessment of functional cognitive performance (memory, processing speed and motor praxis). Some of these tests were repeated at age 53 years (1999), with the addition at this age of 4 tests measuring verbal ability (the NART), verbal fluency, prospective memory and delayed verbal memory. The NART is the National Adult Reading Test and allows a link back to related tests in childhood.

## History of cleaning/derivation/documentation

03/2002 - Cleaning of variables [converted from string format, DK codes (? and ??) and blanks recoded to numeric values].

06/2001 - Derivation of cognitive function scores at **age 53** (1999)

09/2014 - Labelling of cognitive function scores at age 53 plus the derivation of VSHT99.

06/2015 - Summary variables WLT99, WLIN99, CANSP99, NART99, NART99R still need to be derived and added to SWIFT

## Variables in this document

### 21 high-confidence matches

| Variable | Label | How it was found |
|---|---|---|
| **Clinical data [51] › Cognitive function [101] › Adulthood cognition [1012] › Envelope test [10134]** | | |
| remem | How well has the study member remembered tasks involved in envelope test- at age 53 years | topsheet field |
| **Clinical data [51] › Cognitive function [101] › Adulthood cognition [1012] › Memory (word list memory test) [10124]** | | |
| wlcd99 | Word list memory test: Word list card used - at age 53 years | topsheet field |
| wlin199 | Word list memory test: Number of intrusions (Words not on the list) - Test 1 - at age 53 years | topsheet field |
| wlin299 | Word list memory test: Number of intrusions (Words not on the list) - Test 2 - at age 53 years | topsheet field |
| wlin399 | Word list memory test: Number of intrusions (Words not on the list) - Test 3 - at age 53 years | topsheet field |
| wlin499 | Word list memory test: Number of intrusions (Words not on the list) - Test 4 (delayed trial) - at age 53 years | topsheet field |
| wlinx99 | Word list memory test: Number of intrusions (Words not on the list) - Wrong words Test 4 (delayed trial) - at age 53 years | topsheet field |

| Variable | Label | How it was found |
|---|---|---|
| wlt199 | Word list memory test: Number of words recalled (max score 15) - Test 1 - at age 53 years | topsheet field |
| wlt299 | Word list memory test: Number of words recalled (max score 15) - Test 2 - at age 53 years | topsheet field |
| wlt399 | Word list memory test: Number of words recalled (max score 15) - Test 3 - at age 53 years | topsheet field |
| wlt499 | Word list memory test: Number of words recalled (max score 15) - Test 4 (delayed trial) - at age 53 years | topsheet field |
| wlt99 | Word list memory test: Sum of 3 tests (max 45) - at age 53 years | topsheet field |
| wltx99 | Word list memory test: Number of words recalled (max score 15) - Wrong words Test 4 (delayed trial) - at age 53 years | topsheet field |
| **Clinical data [51] › Cognitive function [101] › Adulthood cognition [1012] › National audit reading test (NART) [10126]** | | |
| nart99 | National Adult Reading Test (NART) pronunciation score at 53 years - naturally coded (= number of errors) - at age 53 years | topsheet field |
| **Clinical data [51] › Cognitive function [101] › Adulthood cognition [1012] › Processing speed [10125]** | | |
| cansp99 | Visual letter search: Speed (max 600) - at age 53 years | topsheet field |
| vscl99 | Visual letter search: Column reached at last mark - at age 53 years | topsheet field |
| vser99 | Visual letter search: Errors - at age 53 years | topsheet field |
| vsht99 | Visual letter search: Hit targets - at age 53 years | topsheet field |
| vsms99 | Visual letter search: Number of targets missed - at age 53 years | topsheet field |
| vsrw99 | Visual letter search: Row reached at last mark - at age 53 years | topsheet field |
| **Clinical data [51] › Cognitive function [101] › Adulthood cognition [1012] › Verbal fluency [10127]** | | |
| anin | Animal test: Number of animals listed - at age 53 years | topsheet field |

## 2 medium-confidence matches

| Variable | Label | How it was found |
|---|---|---|
| **Clinical data [51] › Cognitive function [101] › Adulthood cognition [1012] › Memory (word list memory test) [10124]** | | |
| wlin99 | Word list memory test: Total number of intrusions (Words not on the list) - at age 53 years | prose scan |
| **Clinical data [51] › Cognitive function [101] › Adulthood cognition [1012] › National audit reading test (NART) [10126]** | | |
| nart99r | National adult reading test: Correctly pronounced (reverse coded) - at age 53 years | prose scan |

## 1 low-confidence match

| Variable | Label | How it was found |
|---|---|---|
| **Questionnaire data [55] › Health and medical history [510] › Medical conditions [5101] › Musculoskeletal system conditions [51016]** | | |
| back | In the last 12 months, have you had sciatica, lumbago or severe backache at age 53 years | prose scan |